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Fall 2026 **Course:** COMPSCI 566

Monocular Depth Estimation: Benchmarking and a Defocus-Informed Refinement

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Progress Summary

The full pipeline is implemented and evaluated end to end. Phase 1 is complete: DA-V2 is benchmarked on all 654 NYU-Depth v2 test images at three model scales (ViT-S/B/L) plus the metric indoor checkpoint, and the classical depth-from-focus baseline (modified Laplacian focus measure, parabolic sub-frame interpolation, confidence map, all-focus guided smoothing) runs on rendered focal stacks and all 120 DDFF test stacks. Phase 2 is complete on synthetic data: robust scale fit and regularized local correction run on six scenes with a tuning/held-out split, and the combined system runs qualitatively on real DDFF stacks.

Current Results

NYU-Depth v2 test split (654 images, Eigen crop), DA-V2 across model scales and alignment choices:

Checkpoint	Alignment	AbsRel	RMSE (m)	δ_1
ViT-S	median scale only	0.494	2.349	0.394
ViT-S	least squares	0.062	0.297	0.956
ViT-S	robust IRLS	0.058	0.292	0.958
ViT-B	least squares	0.057	0.283	0.959
ViT-L	least squares	**0.055**	**0.277**	**0.961**
Metric-Indoor-S	none (raw metric)	0.218	0.650	0.670
Metric-Indoor-S	least squares	0.077	0.353	0.936

The affine fit matters: a median scale alone collapses accuracy (δ_1 0.39), which supports modeling $q = s\hat{d} + t$ rather than rescaling alone. Robust IRLS buys a small consistent gain over plain least squares. Scaling the backbone up helps less than expected (S $\rightarrow$ L gains $\approx$0.005 δ_1 at 4$\times$ the runtime on MPS), so we keep ViT-S as the project baseline. The metric indoor checkpoint carries a real scale prior but is biased on NYU (0.218 unaligned); aligned, it still trails the relative models.

Synthetic focal stacks (10 frames, focus 0.9–7 m, thin-lens blur model). Regularization weights were tuned on two scenes (gallery, stairs); the table reports the held-out mean over the other four (corridor, objects, slanted, shelves), metric depth scored without alignment:

Method	AbsRel	RMSE (m)	δ_1
DFF raw	0.233	1.552	0.732
DFF + all-focus smoothing	0.221	1.510	0.702
DA-V2 aligned (oracle)	0.271	1.524	0.548
DA-V2 + scale fit from focus	0.308	1.614	0.459
Refined (ours)	0.259	1.483	0.575
Combined (ours)	**0.206**	**1.284**	0.704

The focus baseline is accurate where texture exists (δ_1 0.89–1.00 on textured regions) but

degrades on flat regions, exactly where the confidence map reads low ($\bar{w} \approx 0.09$–0.25 textureless vs. 0.26–0.66 textured); all-focus guided smoothing recovers part of that loss (textureless AbsRel $0.77 \rightarrow 0.62$ on eval scenes) but can propagate confident edges into large flat regions, a known limitation of normalized convolution. On the calibrated evaluation scenes the refinement beats every baseline in every split (AbsRel 0.246 vs. 0.266 focus, 0.312 scale fit; textured 0.211 vs. 0.224). The *combined* system, which falls back to the focus estimate when calibration is rejected, strictly dominates both components. DA-V2 is weaker than the focus baseline on our synthetic scenes even with oracle alignment — plausible, since the renderer is far from its training distribution — which makes this benchmark a useful stress test rather than a real-data statement.

Difficulties

- **DDFF labels.** The public test archive ships 120 focal stacks but no ground truth (withheld for the benchmark server), and per-frame focus distances are not recorded. We reconstructed focus distances from the refocus geometry in the paper (disparities linear in $[0.02, 0.28]$ px, $z \approx 0.14/d$), verified the frame-to-depth ordering per stack by Spearman correlation against the learned prior (about a third of the stacks we checked prefer reversed order), and limited DDFF to qualitative evaluation. Measured confidences on the real Lytro stacks are low ($\bar{w} \approx 0.01$–0.05); a per-scene confidence normalization added to the fit gets calibration accepted on most stacks, but the numbers cannot be scored without labels.
- **NYU test split sourcing.** Both canonical sources were unreachable (the DenseDepth mirror returns 403 and the NYU host refuses connections); we used a mirror of the same 654-image Eigen test split and verified image count, units, and depth range.
- **Defocus veiling.** In the layered renderer, blurred near-field texture bleeds over flat surfaces, which lifts the focus measure there (the synthetic corridor's flat ceiling reaches $w \approx 0.4$). This is a genuine failure mode of the focus measurement rather than a renderer artifact and it caps how far we can trust w at occlusion boundaries.
- **Rejected fits.** On scenes where the learned prediction anticorrelates with the focus estimate, the scale fit is degenerate ($s \leq 0$); we fall back to the bounded uncalibrated output per the proposal rather than emit bad metric depth.

Changes from Proposal

DDFF 12-Scene moved from a quantitative to a qualitative dataset: the verification the proposal made conditional showed the released test stacks carry no labels. Rendered scenes now carry the quantitative refinement evaluation, split into tuning and held-out sets so reported numbers are not fit on the evaluation scenes. Three additions beyond the proposed method: an all-focus guided smoothing stage on the DFF baseline (normalized convolution in inverse depth, weighted by confidence); a combined system output that uses the refined depth where calibration succeeds and the focus estimate where it is rejected, which strictly dominates both components on the held-out scenes; and a degenerate-fit guard replacing silent bad calibration.

Next Steps

- Finish the DA-V2 model-size and metric-head benchmark table and fold it into the webpage and presentation.
- Extend the DDFF qualitative evaluation to more stacks and, time permitting, phone-captured

focal stacks as a second real source.

- Finish the results figures and tables for the project webpage; keep the presentation deck in step with the final numbers.